Which procurement contracts will fail?

Midterm · Baseline model · Tender Risk ML · Aziz Abdulin

TASK

Binary classification: at the moment the winner is selected, predict whether the contract will be terminated or never signed.

WHY

1.8 million procurements a year: auditors can inspect only a small share and need a ranked list of the riskiest ones.

358,789

procurements with a known outcome (Q1 2024)

6.6 %

of them ended with a failed contract

1 : 14

class imbalance — failures are rare

Data: procurements with a known outcome

Source — Open data of the state procurement portal goszakup.gov.kz (OWS v3 API)

Scope — All procurements published in Jan–Mar 2024 that produced a contract; 5 linked tables joined into one row per procurement

Target — contract_failed = 1 if the contract was terminated or never signed

Features — 20 features known at award time: value, method, competition, timing, subject, region

358,789
procurements

6.6 %
failed contracts

20
features

Chronological split, no shuffling



Train: 1 Jan – 29 Feb · Validation: 1–15 Mar · Test: 16–31 Mar

×10
failure rate from the cheapest to the most expensive value decile (1.5 % → 16.5 %)

9 % vs 3 %
requests for quotations vs single-source purchases

Baseline: class-weighted logistic regression

Preprocessing

Median imputation + missing flags, scaling, one-hot encoding; fitted on train only

Logistic regression

L2 penalty, C = 1, class_weight = balanced, solver lbfgs

Risk score 0–1

Ranks procurements; decision threshold tuned on validation

Training: 266,294 procurements (Jan–Feb) · 20 features → 52 inputs · converged in 75 iterations · 1 second

Why this model

1. Outputs a probability, so procurements can be ranked for inspection
2. Readable coefficients: the model can be checked against the EDA
3. Class weights handle the 1 : 14 imbalance without resampling
4. Trains in 1 s on 266 K rows and works on sparse text features

Not the baseline (yet)

Gradient boosting, transformer: planned for A3, they need a baseline to beat. Kernel SVM, kNN: do not scale to 266 K rows. Naive Bayes: features are strongly correlated.

Metrics: why accuracy is the wrong yardstick

93.4 %

accuracy of a “model” that flags nobody

…and finds 0 failed contracts

With only 6.6 % failures, accuracy rewards doing nothing. Our useful models score lower, about 0.89, because they dare to flag.

ROC-AUC **0.79**

Chance that a failed contract is scored above a successful one. Threshold-free, independent of class ratio.

PR-AUC **0.26**

Average precision over all recall levels. Random = 0.066, so 0.26 is 4× better. Most honest under imbalance.

F1 **0.32**

Balance of precision (0.26) and recall (0.42) at one threshold, tuned on validation only.

Precision@1 % **0.50**

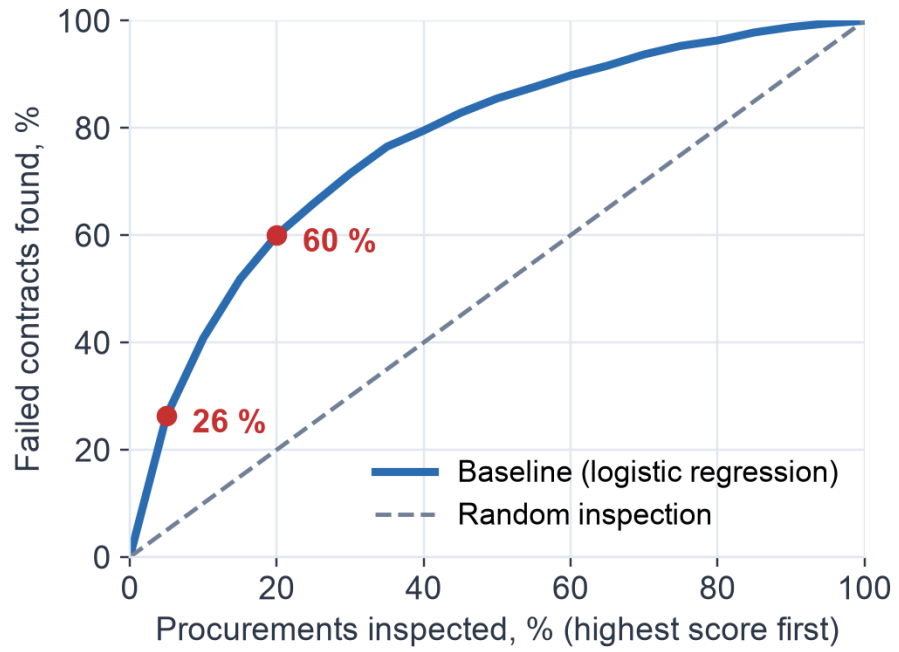
Quality of the top of the list: matches the auditor's limited inspection capacity.

Results: far better than chance on unseen data

Model (test: 16–31 Mar)	ROC-AUC	PR-AUC	F1	P@1 %	R@5 %
Prior (no model)	0.500	0.066	–	0.066	0.050
Logistic regression (baseline)	**0.792**	**0.262**	**0.324**	**0.504**	**0.263**
Random forest	0.804	0.298	0.368	0.563	0.300
LogReg + lot text (TF-IDF)	0.802	0.282	0.332	0.550	0.277

1. Stable: ROC-AUC 0.794 on validation, 0.792 on test
2. Non-linearity matters: random forest gains +0.04 PR-AUC with the same features
3. Lot text adds a little: +0.02 PR-AUC

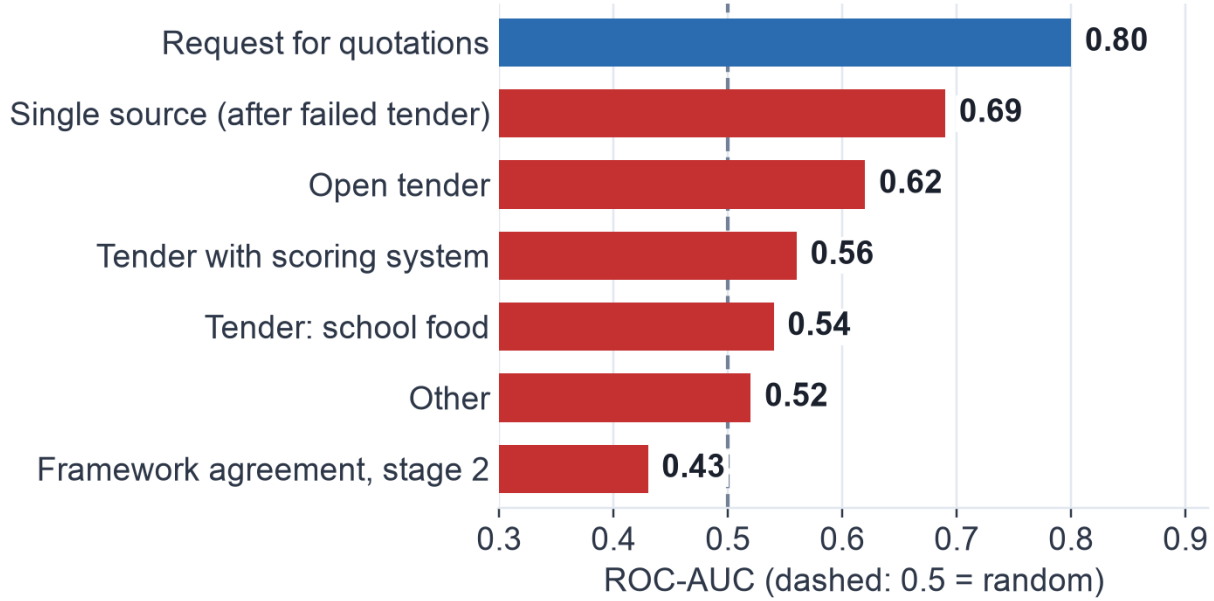
Failures found vs. inspection budget (test)



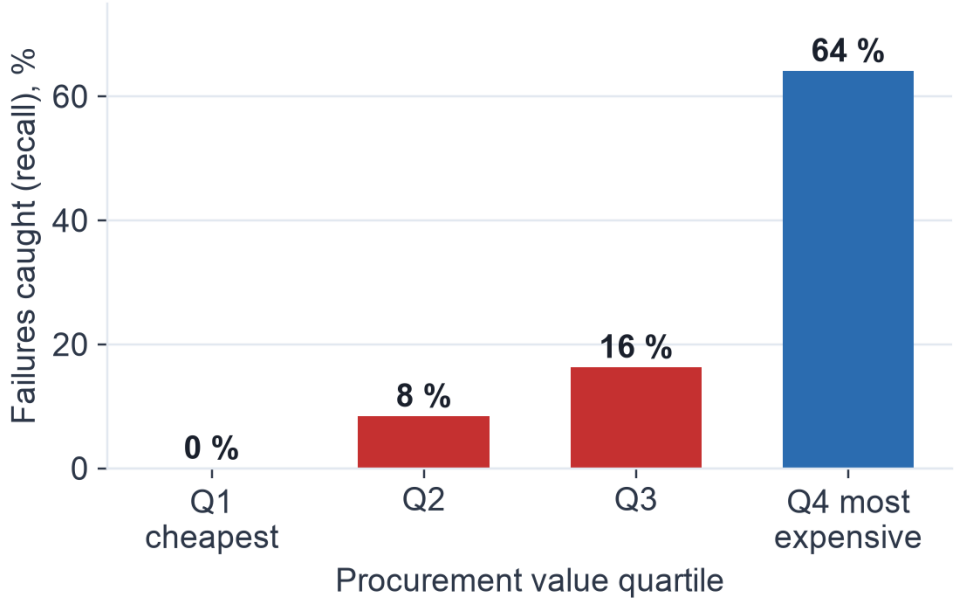
Inspect the top 5 % → find 26 % of failures
Top 20 % → 60 %. At random: 5 % and 20 %

Where the baseline fails

Ranking quality inside each procurement method



Share of failures caught, by procurement value



One method only

Good ranking inside requests for quotations (0.80), near random in the small tender types

Blind to cheap purchases

0 % of failures caught in the cheapest quartile, 64 % in the most expensive

Scores are not probabilities

Class weighting inflates them: mean score 0.42 vs real rate 0.07. Ranking only

Conclusions and next stage

WHAT THE FIRST EXPERIMENT SHOWED

0.79 ROC-AUC on unseen March data: award-time information does predict contract failure

5× better than random inspection: the top 5 % of the list holds 26 % of all failures

0.83 ROC-AUC for the strict failure label: the signal is not label noise

Limits: a linear model, one quarter of data, no supplier track record, uncalibrated scores.

NEXT STAGE · ASSIGNMENT 3

1. Gradient boosting (LightGBM / CatBoost) with hyper-parameter optimisation
2. Probability calibration of the risk scores
3. Fine-tuned multilingual transformer on lot names, combined with the tabular model
4. Per-segment evaluation: non-quotation methods and cheap purchases